Sir Isaac Newton

1642 – 1727

Early Life

He was a curious child, largely raised by his grandmother. He entered Trinity College, Cambridge and probably not well prepared.

The core curriculum was still based on Latin or Greek texts of Aristotle and the various medieval and Renaissance commentators.

He largely were left to fend for himself

Before

Newton began his serious study of mathematics in 1664 during his last year at Cambridge.

At a Cambridge fair, he purchased a book on astronomy which he could not understand because he did not know trigonometry. So he bought a book on trigonometry, which he could not understand because he had never studied Euclid's Elements. So he next purchased Euclid and began what was largely a self-education in mathematics.

After he mastered Euclid, Newton read modern mathematicians, including Kepler and Descartes.

Graduation

Newton graduated in January 1665, but by June he was forced to return to Woolsthrope by an outbreak of the bubonic plague, an epidemic known as the Black Death.

Cambridge University was closed for almost two years until the spring of 1667. During this time in London alone 100,000 out of 450,000 died.

During 18 months away at Home

Invented both differential and integral calculus.

- Calculate orbits by showing what the curvature of the path of an object like the Moon becomes as it constantly changes direction due to the Earth's gravitational field.

Discovered that white light was composed of the colored rays of the spectrum; and,

Found the mathematical law of gravity as it applied to the motion of the moon and bodies on the Earth.

Newtons Job at University

In 1669, when his professor Isaac Barrow resigned, Newton, just 26, succeeded him as Lucasian Professor of Mathematics at Cambridge.

Newton's Differential Calculus

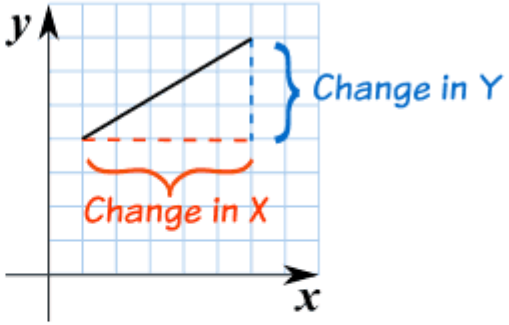
For example, when we drive 50 miles in one hour, we know we averaged a speed of 50 mph. But we also know that we drove faster some of the time and slower some of the time.

We can get the average speed by calculating the difference in position (Δy) and difference in time (Δt) and solving for $\Delta y/\Delta t$.

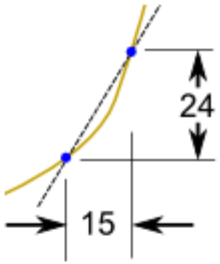
But what do we do when the time interval gets shorter and shorter and then is finally 0 ? - ***Zeno's Paradox***

Slope at a Point

$$\text{Slope} = \frac{\text{Change in Y}}{\text{Change in X}}$$



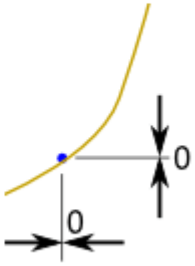
We can find an **average** slope between two points.



$$\text{average slope} = \frac{24}{15}$$

But how do we find the slope **at a point**?

There is nothing to measure!



$$\text{slope} = \frac{0}{0} = ???$$

Derivatives are about Change
Rate of Change

We will use the slope formula:

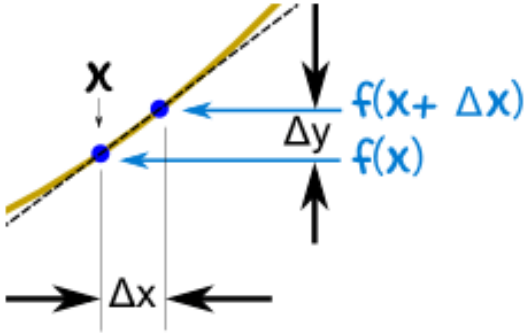
$$\text{Slope} = \frac{\text{Change in Y}}{\text{Change in X}} = \frac{\Delta y}{\Delta x}$$

to find the derivative of a function $y = f(x)$

x changes from x to $x+\Delta x$
y changes from $f(x)$ to $f(x+\Delta x)$

Follow these steps:

- Fill in this slope formula: $\frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$
- Simplify it as best we can,
- Then make **Δx** shrink towards zero.



Derivative Example

Example: the function $\mathbf{f(x) = x^2}$

We know $\mathbf{f(x) = x^2}$, and can calculate $\mathbf{f(x+\Delta x)}$:

Start with: $\mathbf{f(x+\Delta x) = (x+\Delta x)^2}$

Expand $(x + \Delta x)^2$: $\mathbf{f(x+\Delta x) = x^2 + 2x\,\Delta x + (\Delta x)^2}$

The slope formula is:

$$\frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Put in $\mathbf{f(x+\Delta x)}$ and $\mathbf{f(x)}$:

$$\frac{x^2 + 2x\,\Delta x + (\Delta x)^2 - x^2}{\Delta x}$$

Simplify (x^2 and $-x^2$ cancel):

$$= \frac{2x\,\Delta x + (\Delta x)^2}{\Delta x}$$

Simplify more (divide through by Δx):

$$\mathbf{= 2x + \Delta x}$$

And then **as Δx heads towards 0** we get:

$$= 2x$$

Result: the derivative of $\mathbf{x^2}$ is $\mathbf{2x}$

Zenos Pradox Solved

Define the distance across the room as X, and the size of each step as ΔX.

To cross the room in one step: $\Delta X = X$ and $1 \times \Delta X = X$.

To cross the room in two steps: $\Delta X = X/2$ and $2 \times \Delta X = 2 \times X/2 = X$.

To cross the room in three steps: $\Delta X = X/3$ and $3 \times \Delta X = 3 \times X/3 = X$.

To cross the room in n steps: $\Delta X = X/n$ and $n \times \Delta X = n \times X/n = X$.

Zeno challenges us by making the steps smaller and smaller until they become infinitesimal. We must answer the question of what is the value of $n\Delta X$ at the limit when $\Delta X \to 0$? (Note: as $\Delta X \to 0$, $n \to \infty$.)

$$\lim_{\Delta X \to 0} n\Delta X = \lim_{\Delta X \to 0} \{n\,(X/n)\} = \lim_{\Delta X \to 0} \{(n/n)\,X\} = \lim_{\Delta X \to 0} X = X$$

Notice, the key is that $n/n = 1$, no matter how large n becomes. The limit of (n/n) as $\Delta X \to 0$ (or $n \to \infty$) is 1.

1st Law of Motion: Law of Inertia

A body remains at rest or moves in a straight line (at a constant velocity), unless acted upon by a net outside force.

The law of inertia did not originate with Newton, nevertheless it is integral to his system of mechanics. An object in motion will remain in motion unless something acts upon it.

Because a planet is moving in an ellipse (i.e. not a straight line) this law states that there must be some “force” acting upon the planet. If there were no force, the planet would fly off in a straight line.

2nd Law of Motion: F = ma

The acceleration of an object is proportional to the force acting upon it.

The first law says that if no force is acting on an object, it will remain in motion.

The second law tells how the motion will change when a force acts upon the object.

- Velocity is how fast an object is moving (speed or magnitude) and the direction it is moving.
- Acceleration is a change in velocity. An accelerating object can either change how fast it is moving, the direction it is moving, or both.

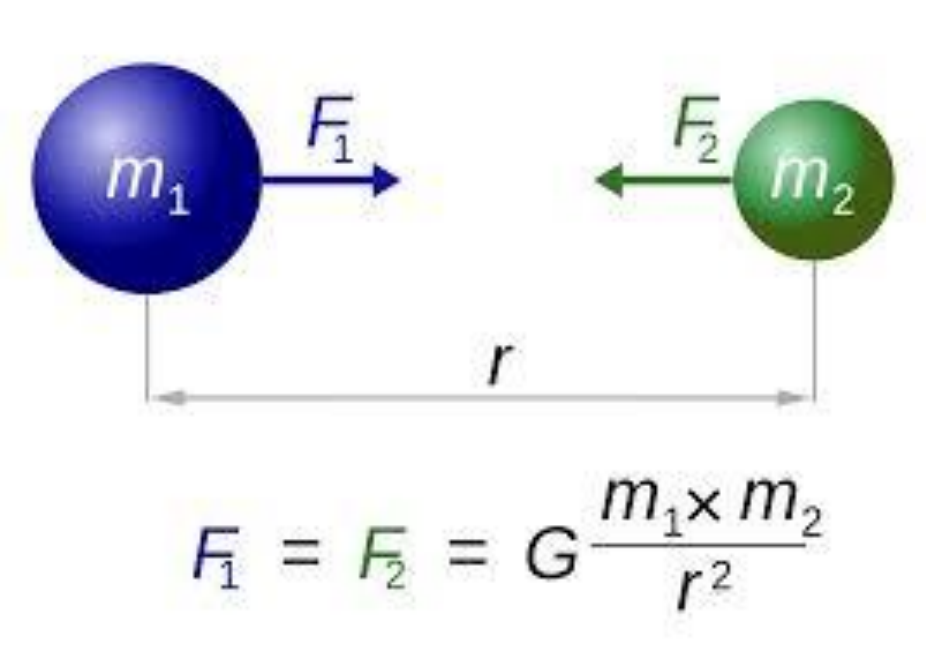
3rd Law: Law of Reciprocal Actions

For every action, there is an equal and opposite reaction.

The law can be more fully stated as, “Whenever one body exerts force upon a second body, the second body exerts an equal and opposite force upon the first body.” That is, when the sun pulls on a planet with the force of gravity, the planet pulls on the sun with a force of equal magnitude. But, because the sun is so much more massive than the planet, Newton's second law says that the sun will experience much less acceleration.

Law of Universal Gravitation

Every object in the Universe attracts every other object with a force directed along the line of centers for the two objects that is proportional to the product of their masses and inversely proportional to the square of the separation between the two objects.



Gravitational Constant: Explained!

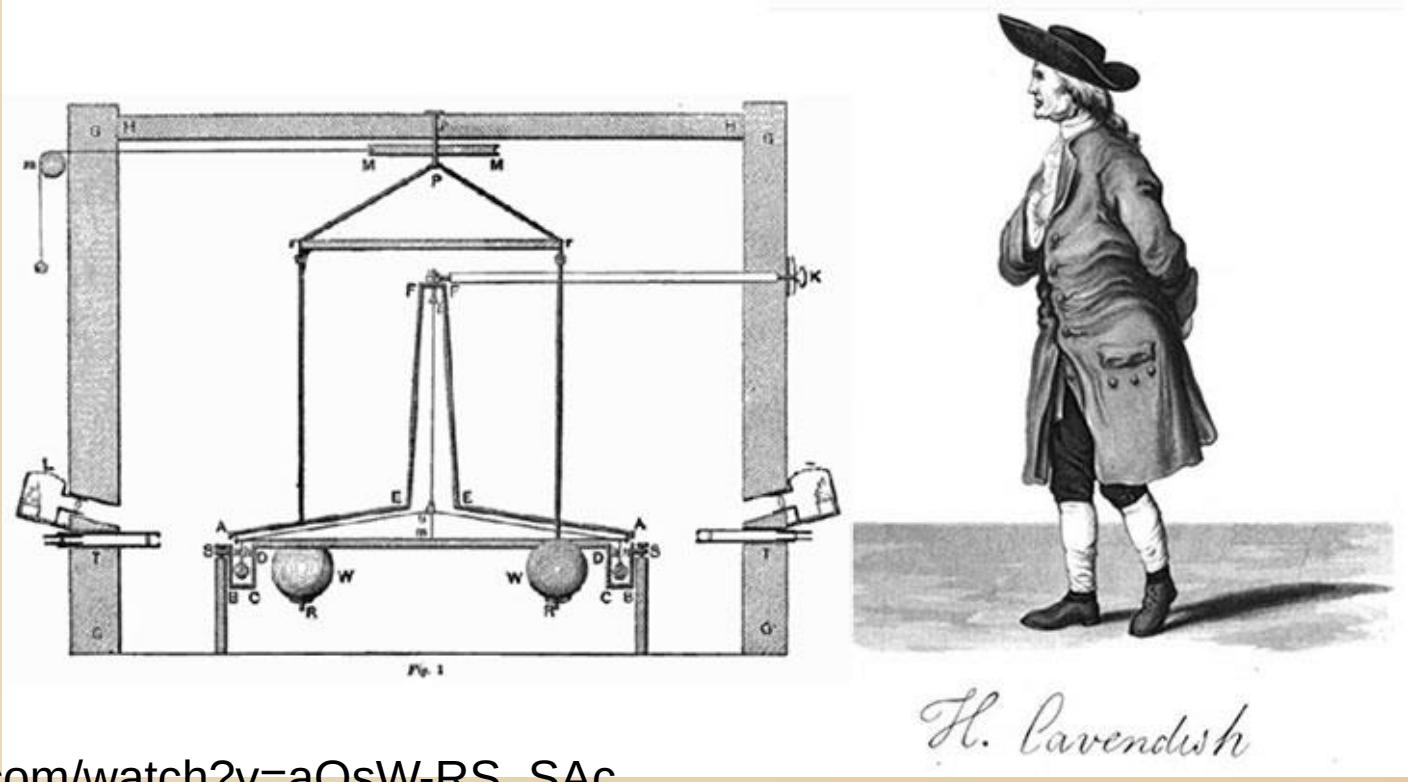
https://www.youtube.com/watch?v=aOsW-RS_SAc

Law of Universal Gravitation

Newton was able to derive all of Kepler's Laws of Planetary Motion from this law and his laws of motion.

$G = 6.67 \times 10^{-11}\ m^3 / (kg\ sec^2)$

Cavendish (1874) calculated using torsion balance. It allowed us to calculate the density of Earth.



- Cavendish Experiment

https://www.youtube.com/watch?v=aOsW-RS_SAc

Mass of Earth - 5.972 × 10^24

1. F = G m M / r^2 = m g

where F is the gravitational force, G is the gravitational constant, M is the mass of the Earth, r is the radius of the Earth,

m is the mass of another object (near the surface of the Earth).

2. G M / r^2 = g (The m's canceled out.)

Now solve for M, the mass of the Earth.

3. M = g r^2 / G,

where g = 9.8 m/sec^2,

r = 6.4 x 10^6 m,

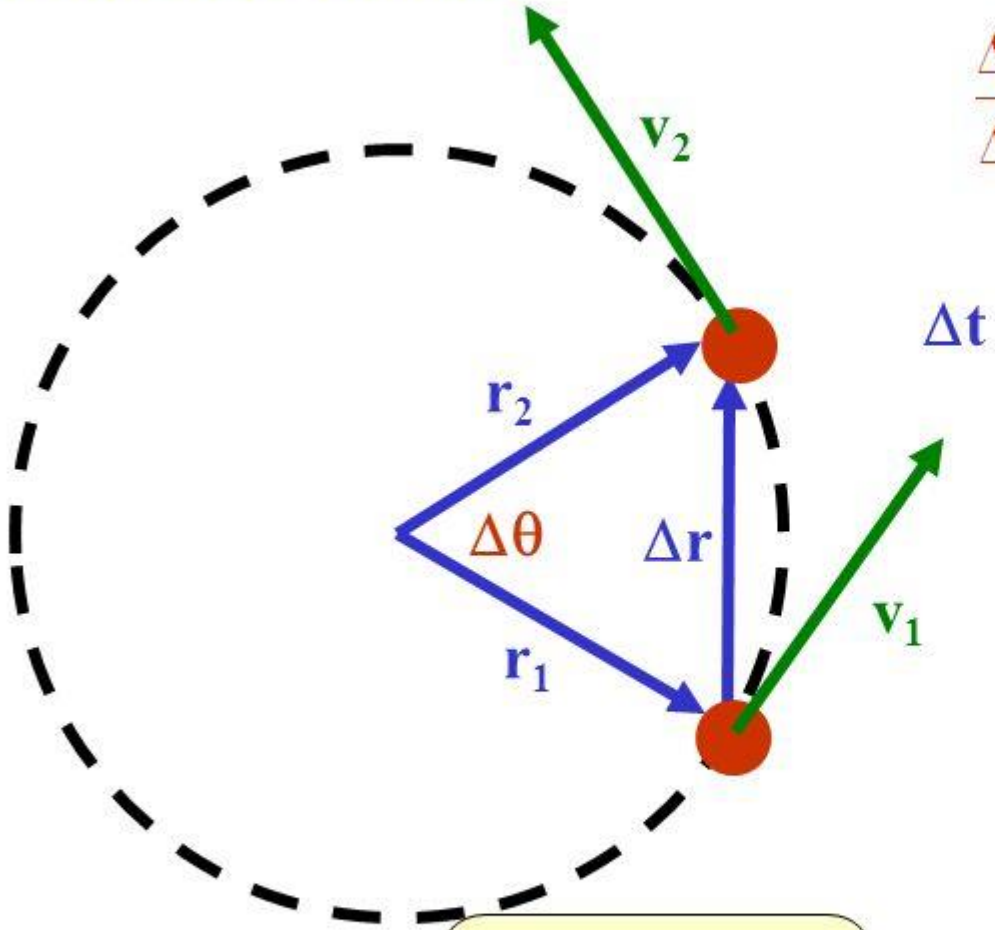
G=6.67 x 10^- 11 m^3/(kg sec^2).

4. M = 9.8 x (6.7 x 10^6)2/6.7 x 10^-11

= 6.0 x 10^24 kg

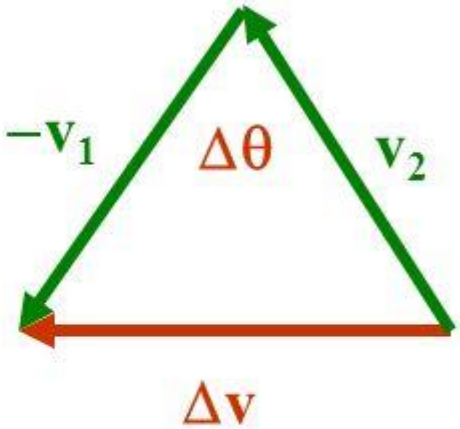
Centripetal Acceleration

Radial Acceleration:



Similar Triangles

$$\frac{\Delta v}{\Delta r} = \frac{v}{r} \quad \Rightarrow \quad \Delta v = \frac{v\Delta r}{r}$$



Divide by time

$$\frac{\Delta v}{\Delta t} = \frac{v}{r}\left(\frac{\Delta r}{\Delta t}\right)$$

Centripetal Acceleration

$$\boxed{a_r = \frac{v^2}{r}}$$

Rotational Motion and the Law of Gravity

https://www.youtube.com/watch?v=XgkzeUxChKU

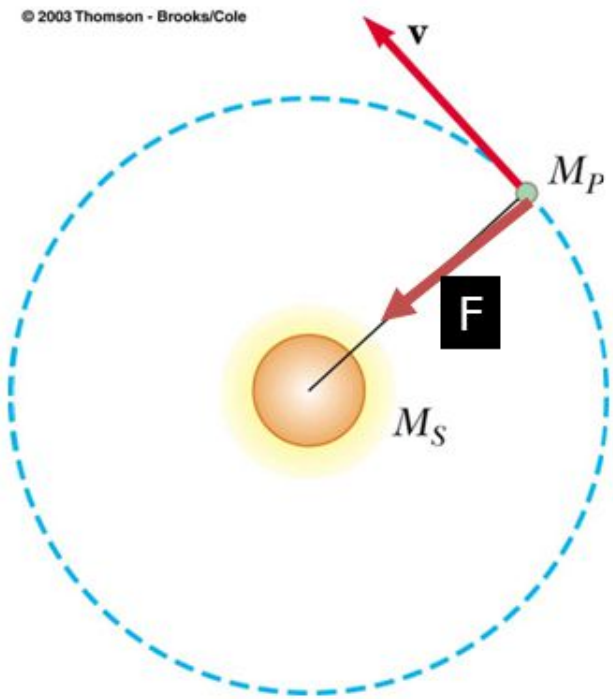
Satellite Law Motion

- Assuming a circular orbit is a good approximation

$$F_g = F_c$$

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$r = \frac{GM}{v^2}$$



V = w r (Angular Speed)

27 days to Circle the Earth

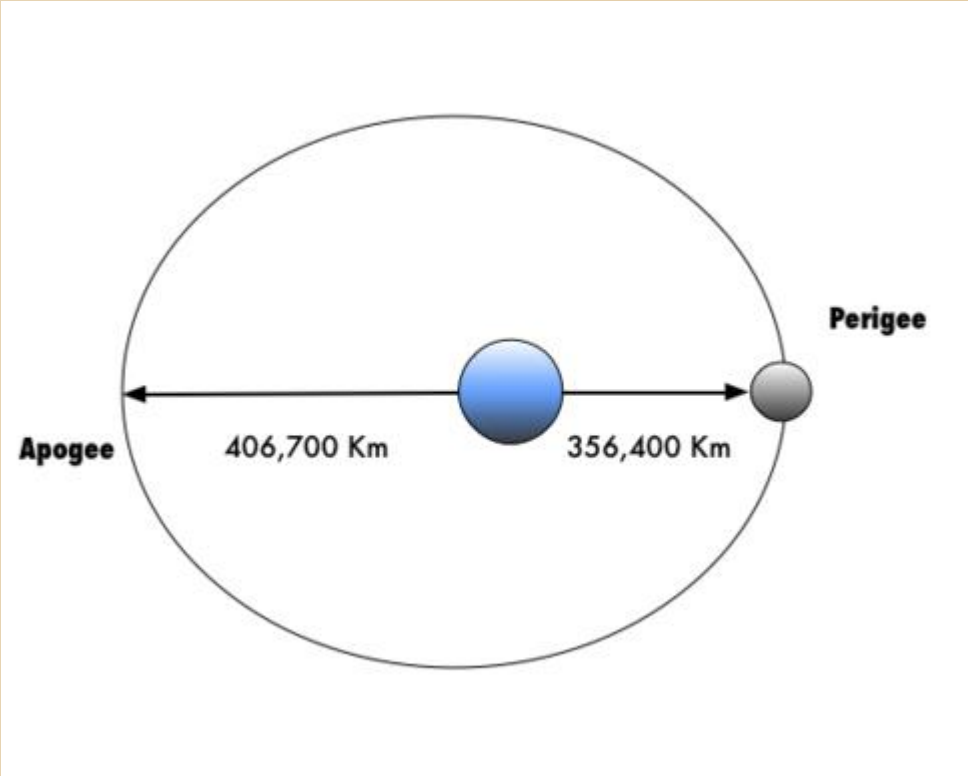
Orbital Move for Moon

Newton’s first law of motion, a body in motion tends to continue in a straight line unless acted upon by an outside force.

The Moon is, however, acted upon by the Earth’s gravity. Gravity accelerates the Moon towards the Earth and causes its path to curve. The combination of these two motions causes the circular motion of the Moon around the Earth

Supermoon

For instance, the size of the Moon can vary by more than 15% from when it's at its closest to when it's at the most distant point.

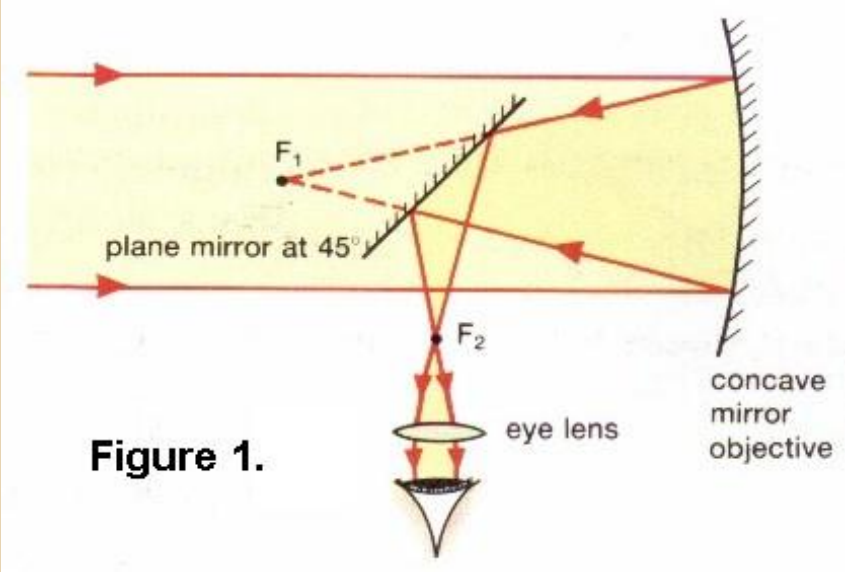


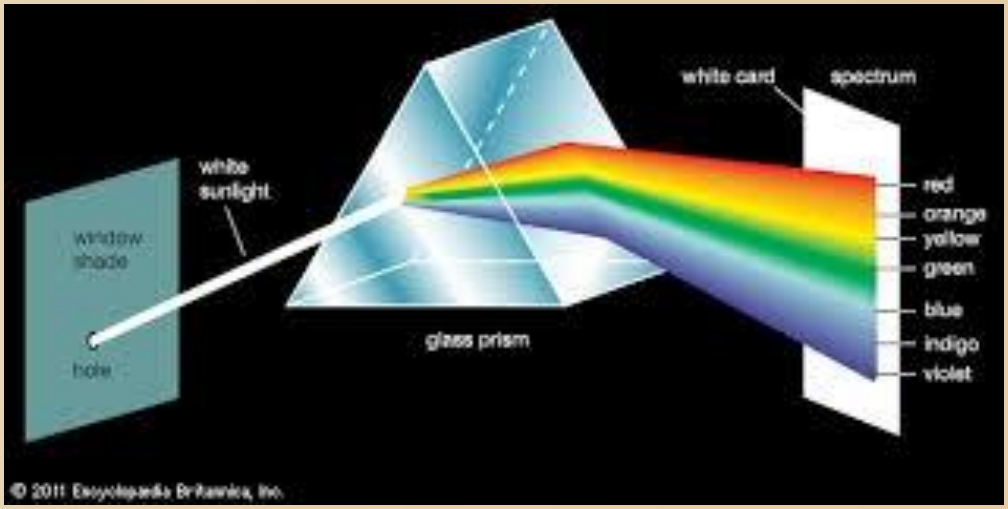
The brightest full Moons occur when the Moon is at the closest, which are typically 30% brighter than when it's farthest away.

Optics

Newton devised the reflecting telescope.

His experiment showing that white light is composite consisted in taking beam of white light and passing it through a prism; the result is a wide beam displaying a spectrum of colors.





Newton's Optics

The color spectrum was explained as weakness of white light. That is, as light moved across the spectrum from white/bright to blue/darkness weakness increases - Robert Hooke

Newton discovered that if a ray of white light passes through a prism at a 45-degree angle, the light is split into the colors of the rainbow.

Each color acts as an independent ray and has its own precise and specific angle of refraction and are not composed of more basic entities.

Newtons Light

According to Descartes the light we see from the sun is simply the result of pressure on the aether. Using the analogy of a blind person's walking stick, Descartes argued that just as the pressure on the end of the stick is transmitted instantly to the hand, so the pressure from the sun transmits light directly to the eye.

Newton believed that this could not be true. For example, at night, he reasoned, one should be able to improve one's vision by running forward to create pressure on the air and aether against the eye. But is does not work so Newton concluded that Descartes was wrong.

Newtons Light

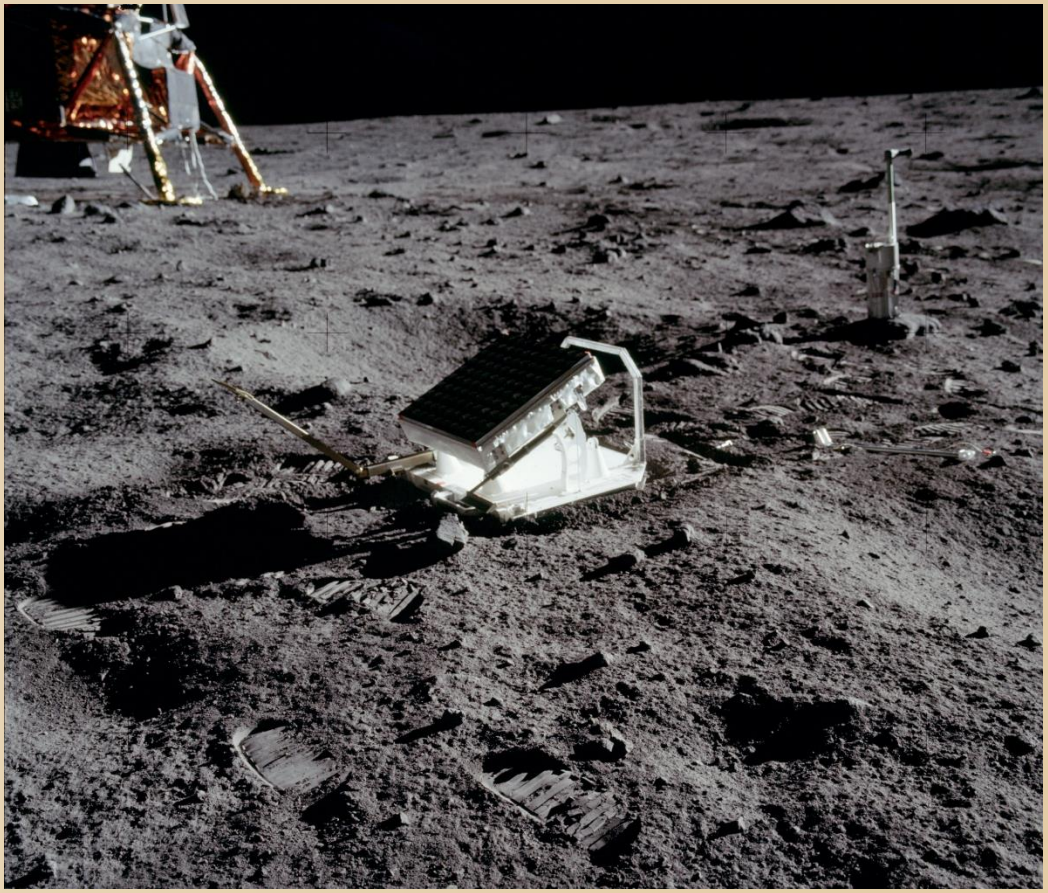
Newton adopted a particulate definition of light, believing that light must travel from the object to the viewer. Newton's transmission theory, of course, assumed that light had a velocity.

Later, as you know, it was observed that light has wave-like qualities.

Distance Between Earth-Moon

Accurate

Today, millimeter-precision measurements of the lunar distance are made by measuring the time it takes for light to travel between LIDAR stations here on the Earth and retroreflectors placed on the Moon.



This process is known as the Lunar Laser Ranging experiment, a process that was made possible thanks to the efforts of the Apollo missions.

Newton’s Principia Mathematica

Aristotle’s theory was, of course, universal but it was based on untenable assumptions such as heavenly bodies being perfect and all orbits being circles. Aristotle’s theory did not agree with scientific observations.

Newton’s Principia restored a comprehensive world-view, which had been lost with the collapse of Aristotelian cosmology and physics.

Newton's Principia Mathematica

The Principia Mathematica stands at the pinnacle of 17 th century science because it: provided the underpinning for Copernicus's system; explained Kepler's laws of planetary motion; affirmed and expanded Galileo's work on terrestrial motion and projectiles; and, completely superseded Descartes' Principia Philosophica.

Principia Mathematica is one of the most important books ever written. It is probably the most important book in science.

Newtons Alchemy

How did inanimate stuff become animate living stuff ?

Where does the vital agent in matter first occur?

Where is spirit? Where is vitality?

Newton believed in the existence of a vital agent diffused through all things.

He spent 30 years trying to find the vital agent and hence a unified theory of creation.

But with no success.

After Newton

Implications of Newton's theory was not apparent immediately. It took several followers to promote his theories.

Voltaire (1694–1778, French) became the most famous of the promoters of Newton. In “Letters Concerning the English Nation” (1733), he devoted four chapters to Newton. Later he published Elements of Newton’s Philosophy (1738).

The Madame du Chatelet (1706–1749), and her “Institutions of Physics” became the most important interpretation of Newtonian mechanics of the time.

Thomas Jefferson (inventor) carried to North America.

After Newton

Laura Bassi (1711–1778) of Italy was also an early Newtonian, and is credited to carrying Newtonianism to Italy.

She was the first woman to be offered a regular teaching post at a European university, and she made Bologna a major center of Newtonian studies and experimentation.

She first taught Optics, and later the Principia.

Newton’s Law in Action

Pierre-Louis Moreau de Maupertuis, first advocated Newtonian theory to the French Academy of Sciences.

Maupertuis in one of the most famous experiments of the 18 th century, traveled to the arctic and the equator and used Newtonian gravitational theory to determine the shape of the Earth.

Scientific Nomenclature

Antoine Lavoisier, the founder of modern chemistry, argued that it is impossible to separate the nomenclature of science from science itself.

Science produces facts; ideas represent the facts; and words express the ideas. Lavoisier concludes: “...we cannot improve the language of any science without at the same time improving the science itself; neither can we...improve a science, without improving the language or nomenclature which belongs to it.”